

Case: Smart Tourist Navigation

Recent surveys from Statistics Norway show 3 million overnight stays at commercial accommodations in September 2022. Around one-third of these visitors stay in Oslo and in Viken county. Although many people come to Oslo for business, we can assume many are interested in tourism in the city. That is the main business of VisitOslo AS, a marketing company that promotes Oslo as a travel destination. Aligned with this goal, VisitOslo also provides IT products and services for the tourism industry, both developed in-house and by agreement with others. Its main product is a web platform (<https://www.visitoslo.com/>) accessible to everyone. Motivated by the increase in visitors, VisitOslo is testing new concepts for IT products, one of which is a Smart Tourist Navigation (STN) app.

The current state: Nowadays, tourists use embedded map applications, such as Google Maps, to navigate the city. It requires the user to input the name of the destination exactly. If a sightseeing place is decided, users do not have any problem. But, if the user wants to explore new places they are unaware of, this system is undesirable. In addition to that, map applications display only the route to specified destinations. To make a sightseeing plan, a tourist must input each visit, choosing the routes for each place separately. A tourist can only visit the places of which they are aware of. Map applications show only the top locations around the tourist. Still, they fail to promote other local attractions and activities related to user preferences.

STN proposal: The purpose of developing this project is to create a mobile application for tourists visiting Oslo that want to explore the city. By specifying the time in hours and locations to be visited, the app smartly establishes a schedule for the tourist based on provided time. STN comprises six modules:

1) Registration and Login: Tourists must first download the mobile app from the Google Store or Apple App Store. Then, they can register their mobile phones by filling out a registration form. STN allows users to register using their Google account or Apple ID. However, it still requires basic user details like email and mobile number. These are sensitive data according to the GDPR, and the company is concerned with data security. The user details will be encrypted and stored in a database administered by VisitOslo. Once the user is registered, the tourist can log in to the application using their Google account or Apple ID.

2) Geolocation: The user is asked to provide permission to access the device's location for the first time using STN. Users can select between 'always' or 'while using the app.' With the help of the device's GPS, which is an inbuilt app, STN can check the current latitude and longitude of the user and compare it to the locations in the VisitOslo database. STN uses Google Maps API to display tourist attractions and activities nearby. The geolocation module requires a working internet connection all the time to work.

3) Creating a visit plan: If the user presses a 'new plan' button, STN asks the user to select the visit's date and time frame (beginning and end time). Once the date and time are captured, users should choose which place they want to visit. Users can search the places in the

VisitOslo database by their name, category (ex., museum), or area (ex., Fornebu). The search result shows a list of places, with a short description, entry fee, distance relative to the user, and opening times. STN will highlight locations with opening times that do not match the inputted time frame, but it allows the user to re-plan it. Users can also set the duration of the visit, and STN suggests a length according to the category (ex., a minimum of two hours for museums).

Once the user selects one desired location, STN traces a route from their current location to the destination. Users can edit the starting location and recalculate the planned route. The places are displayed on a map giving a clear idea of the location and the planned route. The route could include public transportation (ex., train, subway, tram, ferry, or bus) and walking. STN also shows the average time required to reach it (given the starting time and transportation) and the costs for public transit. To provide such information, STN uses Google Maps and Ruter APIs. The total travel plus visit time should not exceed a day.

4) Updating the visit plan: STN also checks for potential tourist attractions or activities close to the planned route (around 200 meters from the route path) and automatically suggests them to the user. Users can filter the suggested locations according to their preferences in categories such as sightseeing, shopping, museums, sports, etc. Users can then select and add the suggested locations in the sightseeing plan. Each time a new place is added, STN updates the traced route, including the travel time and costs for public transportation. STN allows users to change the order and edit the duration of each visit added.

5) List all visit plans: Every time the user creates or updates a visit plan, STN saves it on the mobile. STN lists all visit plans created based on the user selection. Users can access the visit plans to check the schedule on a given day. They can also edit or delete the visit plan if they wish. Finally, users can share their plan with other users of the STN platform or as a post on social media (i.e., Facebook, Twitter, Instagram, etc.).

6) Notifications: STN also has a notification feature that reminds users of the time and the place they are supposed to visit next. This helps users complete the trip according to what they have planned. Once the user creates or updates a visit plan, an alarm with a notification will be set. Once the beginning time matches the device time, the user will be notified that they should start moving.

STN is to be developed for Android and iOS. VisitOslo plans to test an STN prototype containing the main features during the Norwegian National Day in May/2023. The scope of the main features are to be decided by a product owner. The project will evolve the app, including other desirable features, during the remainder of 2023. The official release is on 1st January 2024. A Norwegian IT consulting company has conducted a feasibility study of the project idea and helped VisitOslo estimate its costs. The project budgeted cost is NOK 4 million, and VisitOslo is willing to assume a high-risk overrun up to 25% higher.